SIM2001 Advanced Calculus : Tutorial 1

The Sandwich Theorem. If $g(x,y) \le f(x,y) \le h(x,y)$ for all $(x,y) \ne (x_0,y_0)$ in a disk centered at (x_0,y_0) and if g and h have the same finite limit L as $(x,y) \to (x_0,y_0)$, then

$$\lim_{(x,y)\to(x_0,y_0)} f(x,y) = L.$$

1. Find the limits.

   (13.2.17) $\displaystyle\lim_{\substack{(x,y)\to(0,0)\\ x\ne y}} \frac{x-y+2\sqrt{x}-2\sqrt{y}}{\sqrt{x}-\sqrt{y}}$

   (13.2.21) $\displaystyle\lim_{(x,y)\to(0,0)} \frac{\sin(x^2+y^2)}{x^2+y^2}$

   (13.2.23) $\displaystyle\lim_{(x,y)\to(1,-1)} \frac{x^3+y^3}{x+y}$

2. (13.2.59)

   Does knowing that
   $$1 - \frac{x^2y^2}{3} < \frac{\tan^{-1} xy}{xy} < 1$$
   tell you anything about
   $$\lim_{(x,y)\to(0,0)} \frac{\tan^{-1} xy}{xy}?$$
   Give reasons for your answer.

3. (13.2.61)

   Does knowing that $|\sin(1/x)| \le 1$ tell you anything about
   $$\lim_{(x,y)\to(0,0)} y \sin\frac{1}{x}?$$
   Give reasons for your answer.

4. Find the limit of f as $(x,y) \to (0,0)$ or show that the limit does not exist.

   (13.2.65) $f(x,y) = \dfrac{x^3-xy^2}{x^2+y^2}$

   (13.2.67) $f(x,y) = \dfrac{y^2}{x^2+y^2}$

   (13.2.69) $f(x,y) = \tan^{-1}\left(\dfrac{|x|+|y|}{x^2+y^2}\right)$

5. Find $\partial f/\partial x$ and $\partial f/\partial y$.

   (13.3.13) $f(x,y) = e^{x+y+1}$

6. Find f_x, f_y and f_z.

   (13.3.27) $f(x,y,z) = \sin^{-1}(xyz)$

7. Find the partial derivative of the function with respect to each variable.

   (13.3.37) $h(\rho,\phi,\theta) = \rho\sin\phi\cos\theta$

8. Find all the second-order partial derivatives of the functions.

   (13.3.43) $g(x,y) = x^2y + \cos y + y\sin x$

(13.3.49) $w = x\sin(x^2y)$

9. Verify that $w_{xy} = w_{yx}$.

   (13.3.57) $w = xy^2 + x^2y^3 + x^3y^4$

10. Use the limit definition of partial derivative to compute the partial derivatives of the functions at the specified points.

    (13.3.63) $f(x,y) = 1 - x + y - 3x^2y, \quad \dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$ at $(1,2)$

    (13.3.65) $f(x,y) = \sqrt{2x+3y-1}, \quad \dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$ at $(-2,3)$

11. (13.3.67) [(13.3.75) in 15th Ed.]

    Let $f(x,y) = 2x + 3y - 4$. Find the slope of the line tangent to this surface at the point $(2,-1)$ and lying in the

    (a) plane $x = 2$

    (b) $y = -1$.

12. (13.3.101)

    Let $f(x,y) = \begin{cases} \dfrac{xy^2}{x^2+y^4}; & (x,y) \neq (0,0) \\ 0; & (x,y) = (0,0) \end{cases}$.

    Show that $f_x(0,0)$ and $f_y(0,0)$ exist, but f is not differentiable at $(0,0)$

13. Express dw/dt as function of t, by using the Chain Rule. Then evaluate it at the given value of t.

    (13.4.5) $w = 2ye^x - \ln z, \quad x = \ln(t^2+1), \quad y = \tan^{-1} t,\ z = e^t; \quad t = 1.$

14. Express $\partial z/\partial u$ and $\partial z/\partial v$ as functions of u and v both by using the Chain Rule. Then evaluate them at the given point (u,v).

    (13.4.7) $z = 4e^x \ln y, \quad x = \ln(u\cos v), \quad y = u\sin v, \quad (u,v) = (2,\pi/4)$

15. Find the values of $\partial z/\partial x$ and $\partial z/\partial y$ at the points.

    (13.4.31) $z^3 - xy + yz + y^3 - 2 = 0, \quad (1,1,1)$

    (13.4.33) $\sin(x+y) + \sin(y+z) + \sin(x+z) = 0, \quad (\pi,\pi,\pi)$

16. (13.4.37)

    Find $\partial w/\partial v$ when $u = 0$, $v = 0$ if $w = x^2 + \dfrac{y}{x}$, $x = u - 2v + 1$, $y = 2u + v - 2$.

17. (13.4.39)

    Find $\partial z/\partial u$ and $\partial z/\partial v$ when $u = \ln 2$, $v = 1$ if $z = 5\tan^{-1} x$ and $x = e^u + \ln v$.